

# Biases, Fairness, and Implications of Using AI in Social Media Data Mining

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**Abstract**—Online social media (OSM) has become an integral part of an individual's daily life. The extensive computational power and decision-making ability of artificial intelligence (AI) and the proliferation of user-generated data on OSM have made the opinion one of the key emerging research areas. However, the ease of accessing, manipulating, and mining such user-generated data raises concerns about privacy and security, data and algorithmic biases and fairness. Nevertheless, their personal and societal implications are barely addressed. In this paper, we discuss the limitations, fairness, and biases introduced in data mining and the AI model development. Moreover, we describe the possible implications of using AI systems on users' privacy and address future research directions to mitigate potential biases.

**Index Terms**—online social media, cyberworld, trust, artificial intelligence, machine learning, bias, fairness, privacy

## I. INTRODUCTION

The growing popularity of online social networks (OSNs) as important communication and sharing platforms have significantly contributed to the data analysis and decision-making process in various domains. Since different OSN platforms are widely used to share stories, content, and opinions, a large amount of data is leveraged by artificial intelligence (AI) and data mining researchers to solve numerous real-world problems [20]. Obtained insights can be used to better understand human-behavioral patterns, peoples' opinions about products and services, emotions and sentiments from users' posts, signs of mental illness and depression among others.

Opinion mining (OM) and emotion detection (ED) from social media text data have become an increasingly popular research area within the machine learning (ML) and natural language processing (NLP) research community [28]. Decision-making, social relationships, and behaviors are largely driven by emotions. While emotion detection from facial expressions [40], gait [37], and others has been widely explored, text-based emotion detection is an emerging research area [43]. Twitter is one of the most widely used online social media (OSM) platforms to obtain user-generated text data for developing AI models. Using Twitter's built-in APIs, data acquisition is possible without notifying the users [16]. However, such content could contain one's private and personal information. Therefore, the AI systems having access to users' personal data and making predictions have been often perceived as invasive [5]. Moreover, the social media data mining workflow could also introduce fairness issues and biases in the collected data. The possible reasons [36] depend on 1) who is collecting and annotating the data, 2) what are the demographics of the users

and how diverse the data is, 3) the purpose of collecting the data, etc. Ignoring such factors while acquiring the data could propagate biases in the ML models.

Therefore, addressing the risks associated with the AI systems built on OSM data is crucial. It is essential for technologists to consider users' requirements, expectations, and feedback before using their data in modeling AI systems. Therefore, the primary objective of this paper is to explicitly address the concerns related to the limitations, biases, fairness, and risks associated with different phases of an OSM data mining and AI model development followed by suggesting potential strategies to mitigate the biases.

## II. LITERATURE REVIEW

Sentiment analysis and emotion detection from user-generated social media texts has gained significant attention in the research community. Badugu et al. [6] utilized rule-based approaches to detect user's emotion from tweets using Sentiment140 dataset [18]. Almanie et al. [3] trained an ED model based on the manually annotated tweets consisting of over 4000 Arabic words including emojis. Other research showed how traditional ML-based ED models trained on the ISEAR dataset obtained improved performances compared to the baseline methods [45], [4]. In these works, although the authors acknowledged the experimentation limitations regarding feature extraction or model selection, no concerns related to the ethical aspects were raised. Many researchers extended the ED model for depression and mental illness detection [46], [41]. In these works, the authors explored different linguistics, behavioral and psychological features to train traditional ML models to detect depression, behavioral disorders, and mental illness. Although as a positive sign, Coppersmith et al. [10] considered users' age and gender while preparing the dataset for mental illness detection, various ethical and privacy issues were not addressed by most of the works. Rather, the systems were developed assuming only the net positive outcomes for the target task.

Research on utilizing individuals' personality traits from their interactions with OSM for user recognition has gained significant popularity [26], [27]. Hassanein et al. [22] performed text semantic analysis to predict users' personality traits through their Facebook status updates. Han et al. [21] presented a novel interpretable personality recognition (PR) model based on a personality lexicon. In [15], the authors

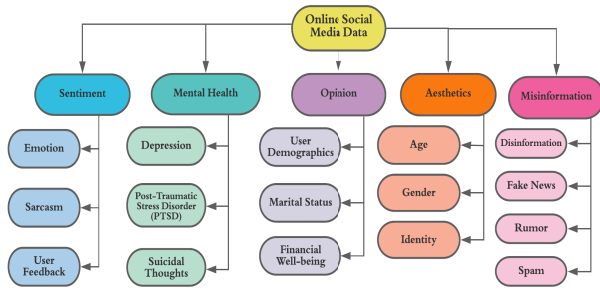


Fig. 1. Taxonomy of online social media data mining

performed YouTube-based PR using different multiple regression techniques. Liu et al. [31] used aesthetic and facial images from OSM for PR. Most of the works did not address ethical concerns. Rather, the primary focus was to enhance the prediction performance and introduce novelty. However, in the documentation of Google’s recently published GoEmotions dataset [12], the limitations and potential biases were explicitly addressed. Such information would keep the other researchers well informed about these limitations. AI is also used to detect misinformation (i.e., disinformation, fake news, rumor, spam) using OSM data [32]. The taxonomy of OSM data mining is presented visually in Fig. 1.

In a nutshell, ML and NLP researchers working on social media data rarely address the limitations, bias, fairness, and implications of their research workflow and outcome. Therefore, in this paper, we aim to discuss these issues and outline strategies to mitigate them.

### III. BIAS AND FAIRNESS: AI IN SOCIAL MEDIA

A traditional ML model development workflow using OSM data is shown in Fig. 2. Generally, the ML model could be an ED model, depression detection model, and others. The first few steps in the OSM data mining involve data collection, annotation, preprocessing, and feature extraction. At first, the raw data is collected followed by performing the data annotation. In the case of Twitter, a dataset can be accessed using Twitter APIs. For ED, existing emotion models such as Paul Ekman’s model or Robert Plutchik’s model [8] can be used to annotate the Tweets. Data can also be annotated using hashtags, emotion keywords, or emojis [25], [35], [50]. Similarly, for personality trait identification, OSM data can be labeled by 16 distinct personality types across 4 axes such as Introversion (I) - Extroversion (E), Intuition (N) - Sensing (S), Thinking (T) - Feeling (F), Judging (J) - Perceiving (P) [8]. OSM dataset prepared for depression detection can be annotated as depressed and non-depressed [44]. After annotating the data, data preprocessing is performed that typically involves tokenization; removal of stop words, hashtags, HTML tags, retweets; spelling corrections, etc. [1]. Once the data have been cleaned and preprocessed, features are extracted using algorithms such as Bag-of-Words, Word2Vec, FastText, GloVe, TF-IDF, etc. [9], [24]. After the data is prepared and the features are extracted, ML models are trained on the training

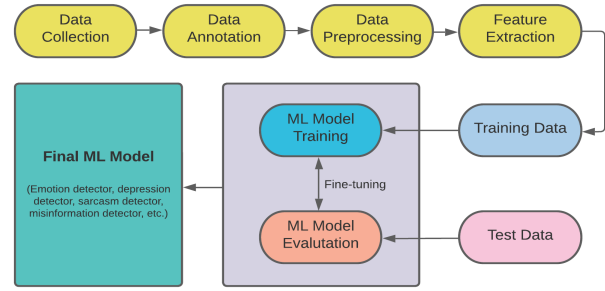


Fig. 2. Different phases of AI in OSM Data Mining

data and evaluated on the test data. Based on the performance of the model, different parameters and hyperparameters are fine-tuned to enhance the classification performance.

#### A. Understanding Bias:

In AI, bias is often considered as the “inclination or prejudice of a decision made by an AI system which is for or against one person or group, especially in a way considered to be unfair” [38]. The significant aspect of the ML system is the training data. Prediction of an ML model primarily depends on the quality (i.e., amount of data, data distribution) of the data used to train the model. Biased data favors some findings, and de-prioritizes or excludes others. Depending on who is collecting and annotating the data, for what purpose the data has been collected, what factors have been considered while collecting the dataset, and many more, there are chances that bias is included in the data. Therefore, while training an ML model, it is crucial to monitor if the amount of data is sufficient to train the model and if the data includes diverse users.

#### B. Ethical Concerns in the Workflow:

In AI-based systems, although developing the overall performance is important from a technological point of view, it is important to consider the overall workflow’s acceptance, reliability, and fairness. Therefore, we discuss the followings:

##### 1) What are the potential biases in OSM data mining?

The OSM data that are used to build the AI models often do not include information related to user demographics. Different demographic information such as gender, age, race, ethnicity, and others could be extremely divisive and bear a heavy historical, cultural, social, or political influence in the users’ overall characteristics. Research shows that people in different cultures would experience emotions differently [11]. Similarly, expression of depression can vary in different cultures [47], age groups [34], countries [30], genders [39]. Depending on the socio-cultural context, individuals would regulate their behavior, emotions, and thoughts in different ways, which is likely to be reflected in their OSM posts as well [29]. Moreover, a sentence could convey different thoughts and emotions in different contexts [19]. Furthermore, someone’s writing proficiency and the way of structuring sentences could be driven by their level of education or first language [13] that could also inform users’ certain behavioral characteristics.

However, the above-mentioned factors are barely considered to prepare the OSM dataset that is used for AI model development. Bias could be introduced in the data annotation phase as well. Manual data annotation can create biased data due to the annotator's demographics, first language, age, education, gender, or political orientation [2]. Annotated data can be biased also due to ambiguous keywords or the lack of linguistic information. For example, Saravia et al. [42] published a dataset to develop an ED system where the tweets were labeled using hashtags. However, in the description of the dataset, it has not been addressed how a tweet has been annotated if it had multiple emotion hashtags (i.e., #happiness and #joy), which is an unaddressed limitation of this dataset.

## 2) Are the predictions of AI models always fair?

An ML model learns what it is been trained on. If the training data is biased, the ML model that learns from the historical data would also be biased [48]. Therefore, after deploying such biased ML models in real-world systems, the models would fail to find distinctive patterns in the dataset and produce false and biased predictions [33]. Such false predictions could negatively impact someone's personal, professional and social life. For example, an ML model using OSM data to monitor users' depression might seem helpful. However, an incorrect prediction about someone's mental state made by a biased ML model could have negative impacts on the users' promotion prospects or increase their insurance premiums. Therefore, it is crucial to remove biased data in the data collection phase that in turn would remove biases from the ML models as well.

## 3) What might be the potential risks and implications of using OSM data to develop AI models?

Given the above-discussed limitations, the assumption of using AI in OSM having a net positive impact should be reconsidered. There are many technological innovations that are praised for solving complex problems with high-efficiency [23]. However, the implications of technology on society are often unaddressed. Since the AI models deal with the data to make predictions about a person's psychological and mental states, one of the crucial considerations should be maintaining the privacy and security of the humans who produce the data. Research also shows that AI systems having access to users' OSM personal data may be perceived as an intrusion of privacy since they have an access to personal, and vulnerable data [5]. As a result, some people might share less about themselves, become fake, and lose connectivity on social media.

## IV. BIAS MITIGATION STRATEGIES

Based on the identified reasons for potential biases in various phases of OSM data mining and AI model development, the following recommendations can be made to mitigate them.

### A. Introducing Diversity in the Dataset

One of the prime factors of having bias and fairness issues is to train the AI model with a non-diverse dataset. Since users' behavior, opinions, and emotions can vary in different regions, the AI model trained on a dataset collected from North America might not generalize to the South Asian

dataset. Including users representing diverse demographics, age, country of origin, writing proficiency, and others would add diversity to the data. As a result, while developing an AI model for OSM data, the extracted features would be more representative and the prediction of the models would generalize better. It recently demonstrated promising results for facial expression emotion detection [14]. Therefore, one of the potential future research directions could be investigating different methods of adding diversity to the dataset.

### B. Adding Datasheets for Datasets

As proposed in [17] and [7], OSM researchers could adopt the practice of introducing datasheets for datasets. The method of data collection and annotation, potential limitations, fairness, and biases need to be explicitly addressed in the documentation of the publicly available datasets. It would inform future researchers about the purpose of the dataset, potential ethical concerns, and so on [12]. Hence, a potential future research direction could be exploring effective ways of including comprehensive datasheets with the datasets.

### C. Qualitative Data Analysis

It is often assumed that the abundance of data alone can help us build the most optimal AI system, which is not always true. In [49], Wang et al. introduced the term "thick data" – the unquantifiable and precious data collected from human experience and narratives, that comes in the form of a very small sample size, but delivers an incredible depth of insights. Therefore, using qualitative research, the researchers could identify people's attitudes towards and acknowledge human complexities. It does not necessarily imply building more accurate technologies to infer these complexities. Instead, people's preferences, expectations, and values should be addressed and prioritized by the innovations. Hence, an exciting avenue for future research would be accounting for human attitudes and the social contexts in designing such technologies that could result in developing fair, accurate, and reliable models.

## V. CONCLUSION

In conclusion, this paper created a systematic classification of research in OSM data mining domain, provided a comprehensive overview of the advancement of AI in social media data mining and discussed its limitations; potential biases, fairness, and privacy concerns associated with the overall workflow. It then presented potential strategies to mitigate the data and algorithmic biases and outlined the future direction of investigations in this research field.

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